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Table of contents

Preface	3
1 Introduction	4
1.1 Use Case & Motivation	4
2 Data Source Description	5
2.1 Overview	5
2.2 Data Sources	5
2.3 Data Mart Architecture	5
2.3.1 Table Relationships	5
2.4 Table Definitions	6
2.4.1 Tax Account Table	6
2.4.2 Appraisal Account Table	9
2.4.3 Improvement Table	14
2.4.4 Improvement Built-As Table	18
2.4.5 Improvement Detail Table	23
2.4.6 Land Attribute Table	24
2.4.7 Sale Table	25
2.4.8 Seg/Merge Table	29
2.4.9 Taxpayer Table	30
2.4.10 Tax Description Table	30
2.5 Key Relationships	31
2.6 Data Quality Notes	31
2.7 Data Dictionary Notes	31
2.8 Access & Contact	32
3 Summary	33
References	34

Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Introduction

1.1 Use Case & Motivation

Course Objectives: • In groups carry out in R the entire data science process on a concrete use case. • Using a reference dataset, demonstrate the process from data acquisition to the development of a prototype solution. • Presentation and documentation of the solution engineering.

The project presentation should include the following points: • Project objective (qualitative and quantitative objectives) [4 points] • Description of the data source (reference dataset) [4 points] • Project organization and allocation of tasks [4 points] • Project development plan (workflow), backlogs [4 points] • Answering

Use case presentation, problem definition, motivation, objective [5 points] What are your goals? (qualitative/quantitative) What are your deliverables? Timeline, milestone plan, etc.

This is a book created from markdown and executable code.

See Knuth (1984) for additional discussion of literate programming.

2 Data Source Description

2.1 Overview

The Pierce County Assessor-Treasurer Data Mart contains comprehensive property assessment and appraisal data maintained by Pierce County, Washington. The data is organized into multiple related tables that track property characteristics, improvements, valuations, and sales transactions.

2.2 Data Sources

- **Tax and Assessment System:** Maintains current tax account information
- **Appraisal System:** Maintains detailed property appraisals and improvements
- **Public Records:** Available through Pierce County Public Records Request Center

Note

RCW 42.56.070(9) may restrict the release of taxpayer names and mailing addresses in certain circumstances. Contact the Pierce County Public Records Request Center for additional information.

2.3 Data Mart Architecture

2.3.1 Table Relationships

The data mart uses a relational schema with the following structure:

Tax Account (Parcel Number)

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Appraisal Account (Parcel Number)

 Tax Description (Parcel Number, Line Number)

Land Attribute (Parcel Number, Attribute)
 Improvement (Parcel Number, Building ID)
 Improvement Built-As (Parcel Number, Building ID, Built-As Number)
 Improvement Detail (Parcel Number, Building ID, Detail Type)
 Seg/Merge (Seg/Merge Number, Parcel Number)
 Sale (ETN, Parcel Number)

Taxpayer (Parcel Number)



Key

Field names in **bold** represent primary key(s).

2.4 Table Definitions

2.4.1 Tax Account Table

Tracks current tax and assessment account information for each property.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Account Type	Text	5	REAL - Real Property; STRUC - Structures; PERS - Personal Property; MOBIL - Mobile Home

Field Name	Type	Size	Description
Property Type	Text	5	ASIMP - Administrative Seg Improvement; LNDIM - Land and Improvements; MBLHM - Mobile Home; RRLEA - Railroad Leases; SAPP - State Assessed Personal Property; SARP - State Assessed Real Property; STRUC - Leased Land and/or Structure
Site Address	Text	50	Physical location address of the property. May be different than mailing address. "XXX" indicates estimated address using GIS
Use Code	Text	5	4-digit numeric code for present highest and best use of the property
Use Description	Text	50	Text description of the numeric use code
Tax Year - Prior	Integer		Tax year for which all prior year values apply
Tax Code Area - Prior Year	Text	10	Code describing geographical area and specific combination of taxing districts. Subject to annual change
Exemption Type - Prior Year	Text	40	Type of exemption, if any, applied to the property

Field Name	Type	Size	Description
Current Use Code - Prior Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Prior Year	Integer		Total fair market value of land, as determined by appraisal process
Improvement Value - Prior Year	Integer		Total fair market value of improvements (buildings, structures), as determined by appraisal process
Total Market Value - Prior Year	Integer		Total fair market value of land and improvements, as determined by appraisal process
Taxable Value - Prior Year	Integer		Market value of the property, minus any exemptions granted
Tax Year - Current	Integer		Tax year for which all current year values apply
Tax Code Area - Current Year	Text	10	Code describing geographical area and specific combination of taxing districts
Exemption Type - Current Year	Text	40	Type of exemption, if any, applied to the property

Field Name	Type	Size	Description
Current Use Code - Current Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Current Year	Integer		Total fair market value of land for the parcel
Improvement Value - Current Year	Integer		Total fair market value of improvements (buildings, structures) for the parcel
Total Market Value - Current Year	Integer		Total fair market value of land and improvements for the parcel
Taxable Value - Current Year	Integer		Market value of the property, minus any exemptions granted
Range	Text	10	Range
Township	Text	10	Township
Section	Text	20	Section
Quarter Section	Text	20	Quarter
Subdivision Name	Text	50	Subdivision name
Located On Parcel	Text	10	Real property parcel number on which this improvement-only parcel is located (Personal Property, Mobile Homes, Leasehold Structures)

2.4.2 Appraisal Account Table

Detailed property appraisal data with characteristics and valuations.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Appraisal Account Type	Text	15	Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest
Business Name	Text	50	Main business name on the parcel, or space number for mobile/manufactured home in a park
Value Area ID	Text	10	Identifies the Physical Inspection cycle for each parcel (PI1 to PI6)
Land Economic Area	Text	30	Also known as LEA, ties parcels to land valuation models. For Commercial: identifies land use model by PI year. For Residential: location code identifying appraisal area, sector and subsector. For Res Com Condos: relates directly to account land value
Buildings	Integer		Represents the number of improvements (buildings) located on a parcel

Field Name	Type	Size	Description
Group Account Number	Text	30	Identifies and ties all parcels qualifying as one economic unit (contiguous assessment). Also identifies all parcels in a condominium project
Land Gross Acres	Decimal	15.4	Total acres that make up the economic unit which could include multiple parcels
Land Net Acres	Decimal	15.4	Size of the individual parcel in acres
Land Gross Square Feet	Decimal	15.4	Total square feet that make up the economic unit which could include multiple parcels
Land Net Square Feet	Decimal	15.4	Size of the individual parcel in square feet
Land Gross Front Feet	Decimal	15.4	For commercial parcels: front feet of the site facing the arterial. For residential: front feet of the lot facing the address side. For waterfront: effective waterfront length of individual lot or combination of lots under single ownership

Field Name	Type	Size	Description
Land Width	Integer		Residential Use Only. Equal to the Net Effective Waterfront Front Feet of the individual parcel, or using Group Account allocation method, then the sum of Net Effective Waterfront Front Feet for the group
Land Depth	Integer		Residential Use Only. Equal to the Net Effective Depth of the individual waterfront parcel, or using Group Account allocation method, then the sum of Net Effective Depths for the waterfront group. Measured from the top of the bank (where bulkhead begins) to the inland property line
Submerged Area Square Feet	Integer		Area of a parcel that is submerged. Not included in the Gross or Net Square Feet
Appraisal Date	Date		Indicates the most recent date an appraiser physically observed the property and/or made changes to the record
Waterfront Type	Text	30	Describes the type of waterfront the property adjoins or has legal access to

Field Name	Type	Size	Description
View Quality	Text	30	Assigned to reflect the market appeal of the overall view available from the dwelling or property
Utility Electric	Text	30	Identifies if power is installed, available or is not available on the property
Utility Sewer	Text	30	Identifies if sewer/septic is installed, available or not available if the property does not support an on site sewage disposal system (no perc)
Utility Water	Text	30	Identifies if water is installed, available or is not available
Street Type	Text	30	Identifies if the access street is paved or unpaved. If blank, the parcel could be a reference parcel or a mobile home assessed as personal property
Latitude	Decimal	10.5	The latitude of the parcel centroid. Some parcels may not have latitude/longitude coordinates (building-only, mineral rights and reference parcels, as well as others not represented in the GIS system)
Longitude	Decimal	10.5	The longitude of the parcel centroid

2.4.3 Improvement Table

Detailed building improvement data linked to appraisal accounts.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel. Not all improvements are on a separate record and may not always be in sequence
Property Type	Text	15	Links buildings to proper cost and depreciation tables. Options include: Commercial, Duplex, Industrial, Mobile Home, Multiple Unit, Out Building, Residential, Townhouse, Triplex

Field Name	Type	Size	Description
Neighborhood	Text	10	Represents boundaries for land subject to similar social, environmental, economic and governmental forces. Boundaries are typically located within a geographic area defined by natural, man-made, or political boundaries. Residential Neighborhoods and LEA's are the same. Commercial Neighborhoods define the location of the property while Commercial LEA's define the use of the property

Field Name	Type	Size	Description
Neighborhood Extension	Text	10	Based on property type: Commercial - represents the primary occupancy or use of the property. Residential - not used, therefore they are set to zero. Mobile/Manufactured Homes - based upon the underlying land ownership. All Mobile/Manufactured homes located on land owned by the Mobile/Manufactured home owner are coded with the Neighborhood Extension 0. Exception - Mobile/Manufactured homes located within a park and owned by the park owner, will carry the same Neighborhood Extension as the park. Mobile Homes in a park have a neighborhood extension of: PL 1 (P = Park, L = Leased); P2, P3, P4, etc

Field Name	Type	Size	Description
Square Feet	Integer		Sum of the square feet for the 'built as' types identified for the building. Each building may have one or more 'built as' type (see IMPROVEMENT BUILTAS table), though usually only one. The exceptions are 'built as' # 124 and 126 (Add on only Res & Com) or any type of Storage Tank, where it represents the count of those 'built as' as opposed to the sum of their square feet
Net Square Feet	Integer		Used primarily to calculate income on commercial properties, where it typically represents the sum of the net rentable area(s). This number may be representative of one or more buildings. If part of a Group Account, these buildings may be located on more than one parcel. For Residential it is equal to the built as square feet or zero (0)

2.4.4 Improvement Built-As Table

Detailed specifications for building construction styles and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Built-As Number	Integer		Unique number identifying each Built-As on an improvement. Sequential number assigned by the system starting with the largest Built-As being number 1
Built-As ID	Integer		Number associated with the description of the purpose or style of construction of the building
Built-As Description	Text	255	Original purpose for and/or style of construction of the building. The Built As is associated with the Marshall and Swift cost and depreciation tables

Field Name	Type	Size	Description
Built-As Square Feet	Integer		Sum of the square feet for the 'built as' types identified for the building. Each building may have one or more 'built as' type, though usually only one. The exceptions are 'built as' # 124 and 126 (Add on only Res & Com) or any type of Storage Tank
HVAC	Integer		Code associated with the predominant heating source for the built-as structure
HVAC Description	Text	30	Text description associated with the predominant heating source for the built-as structure (Forced Air, Electric Baseboard, Steam, etc)
Exterior	Text	25	Predominant type of construction materials used for the exterior siding on Residential Buildings
Interior	Text	15	Predominant type of materials used on the interior walls (Sheetrock or Paneling). Collected for Residential and Mobile Homes only
Stories	Decimal	13.2	Number of floors/building levels above grade. Stories do not include attic or basement areas

Field Name	Type	Size	Description
Story Height	Integer		Based on property type: Commercial - determined by taking an average of the overall ceiling heights of all floors for the Blt As. Residential - determined by the majority of the story height for the first floor (main). Story height is automatically rounded by the system (The system rounds up for .5 and higher and rounds down for anything below .5)
Sprinkler Square Feet	Integer		Total square footage covered by a sprinkler system. Sprinkler SF does not always equal the built as square footage
Roof Cover	Text	20	Material used for the roof (Composition Shingles, Wood Shake, Concrete Tile, etc)
Bedrooms	Integer		Number of bedrooms listed for a residential property (Collected for informational purposes only)

Field Name	Type	Size	Description
Bathrooms	Decimal	7.2	Number of baths listed for a residential property. The number is listed as a decimal, i.e. 2.75 = two full and one three-quarter baths. A tub/sink/toilet combination (plus any additional fixtures) is considered 1.0 bath. A shower/sink/toilet combination (plus any additional fixtures) is 0.75 bath. A sink/toilet combination is .5 bath
Units	Integer		Number of separate units within the building. For Commercial properties, Units may be combined to generate an income approach for the economic unit on building one
Class Code	Text	20	Code for one of five basic cost groups by type of framing (supporting beams and columns), walls, floors and roof structures, and fireproofing. Typically Commercial Buildings

Field Name	Type	Size	Description
Class Description	Text	50	Text description for one of five basic cost groups by type of framing (supporting beams and columns), walls, floors and roof structures, and fireproofing. Typically Commercial Buildings
Year Built	Integer		Year the building was built, as stated by the building permit or a historical record
Year Remodeled	Integer		Year in which improvements are made to the existing structure that extend the life of the structure
Adjusted Year Built	Integer		If greater than the Year Built, it indicates improvements have been made to the original property over and above normal maintenance which would effectively reduce the age of the building
Physical Age	Integer		Tax year less the adjusted year built equals the physical age of the built as
Built-As Length	Integer		Length of the mobile/manufactured home

Field Name	Type	Size	Description
Built-As Width	Integer		Width of the mobile/manufactured home
Mobile Home Model	Text	50	Model of the mobile/manufactured home

2.4.5 Improvement Detail Table

Additional specific improvement characteristics and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Detail Type	Text	10	Category that specific improvement characteristics are coded under. Options include: Add On, Appliance, Balcony, Basement, Carport, Elevator, Fixture, Garage, Mezzanine, Porch, Rough In, Storage. The Add On type allows the appraiser to assign a quality different from the main structure for the characteristics within this field

Field Name	Type	Size	Description
Detail Description	Text	50	Brief explanation of the improvement characteristic (Fireplace, PreFab/Stoves, Laundry Facility, Type of Porch (Cvrd Wd Deck), Type of Bath (Bath 2 Fixture))
Units	Decimal	15.4	Component parts of the building characteristics. The units indicate either the square footage of the structure or a count of each component. Examples: Appliance, Allowance, 1 (count) or Porch, Open Slab, 337 (sf), or Fixture – Bath 3 Fixtures – 2 (count)

2.4.6 Land Attribute Table

Land-specific attributes and characteristics affecting valuation.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property

Field Name	Type	Size	Description
Attribute	Text	30	Describes the category that a land attribute is assigned to: Residential (R), Commercial (C) or Res Com Condo (X) and the subcategory, such as Amenities, Economic, Functional, Neighborhood, Utilities, etc
Attribute Description	Text	30	Further defines the land attribute. Land attributes may or may not contribute to value (Examples of attributes: View Avg, NBHD Gated, WF Bank Low, water available etc)

2.4.7 Sale Table

Historical property sales transaction data.

Field Name	Type	Size	Description
ETN	Text	11	Recording number issued by the Auditor's Office when a property transaction is recorded. Excise tax must be paid before the document of conveyance can be recorded or the mobile home title transferred
Parcel Count	Integer		Number of parcels associated with a sale
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Sale Date	Date		Date the legal document (deed) was executed
Sale Price	Decimal	15.2	Dollar amount recorded on the ETN
Deed Type	Text	40	Type of document which conveyed an interest in or legal title to the property
Grantor	Text	80	Individual(s) or company conveying ownership or interest in the property described on the deed
Grantee	Text	80	Individual(s) or company purchasing ownership or interest in the property described on the deed

Field Name	Type	Size	Description
Valid/Invalid	Text	7	Used to code the validity of a sale for appraisal purposes. A sale in the open market between two unrelated parties, each of whom is reasonably knowledgeable of market conditions and under no undue pressure to buy or sell would be considered a valid transaction. However, a sale can be a valid transaction but if the conditions of the sale fit one of the 27 deletion categories identified in the State Ratio RCW, the sale would be coded invalid for Assessor purposes (sale between relatives, estate sale, percent interest, etc)
Confirmed/Unconfirmed	Text	11	Identifies if the property characteristics at the time of the sale and/or circumstances of the sale were verified by Assessor-Treasurer staff

Field Name	Type	Size	Description
Exclude Reason	Text	30	State required description of why a sales transaction must be excluded from sales studies. Examples include non-arms length sales transactions such as “Family different last names”, sale transactions with undue pressure such as “Estate sale” and “Forced Sale Trans in Lieu Fcl”, and sales where the property characteristics have changed since the sale, “Improved after sale”
Improved/Vacant	Text	8	If any parcel in the sale has any building value it is coded as improved. Vacant is vacant land with no building value
Appraisal Account Type	Text	15	How a parcel is classified (Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest)

2.4.8 Seg/Merge Table

Property segregation and merger transaction history.

Field Name	Type	Size	Description
Seg/Merge Number	Text	12	Unique number assigned to each segregation
Parent/Child Indicator	Text	1	P - Parent Parcel - inactivated parcel being replaced by the new child parcel. There may be zero, one or more parent parcels involved with each segregation. In some cases, a parent parcel may remain active after the segregation. C - Child Parcel - a new parcel created as a result of the seg. There are one or more child(ren) parcels involved with each segregation
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Continued Indicator	Text	1	Y - the parcel is still active after the segregation; N - the parcel is no longer active after the segregation
Completed Date	Date		Date that the segregation was completed
Tax Year	Text	4	Tax year of the segregation

2.4.9 Taxpayer Table

Property owner information (subject to privacy restrictions).

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property

Warning

RCW 42.56.070(9) may restrict the release of taxpayer names and mailing addresses in certain circumstances. Please contact the Pierce County Public Records Request Center to obtain additional information and/or initiate a request for the taxpayer table.

2.4.10 Tax Description Table

Detailed tax coding and descriptions for parcels.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Line Number	Integer		Sequential number identifying each line of a tax description
Tax Description Line	Text	200	Tax description text

2.5 Key Relationships

- **One-to-Many:** Each parcel (Parcel Number) can have multiple improvements (Building IDs)
 - **One-to-Many:** Each improvement can have multiple built-as specifications and detail records
 - **One-to-One:** Each parcel has one appraisal account and one tax account
 - **One-to-Many:** Each parcel can have multiple sales transactions (ETN)
 - **One-to-Many:** Each parcel can have multiple land attributes
 - **One-to-Many:** Each parcel can be involved in multiple seg/merge transactions
-

2.6 Data Quality Notes

- **Historical Data:** Seg/Merge and Sale tables contain historical data. All other tables contain data for active parcels only
 - **Missing Coordinates:** Some parcels (building-only, mineral rights, reference parcels) may not have latitude/longitude coordinates
 - **Address Estimation:** If address contains “XXX”, it was estimated using GIS
 - **Appraisal Dates:** Fields like Appraisal Date indicate when the property was last physically inspected
 - **Value Dates:** Land, Improvement, and Total Market Values are tracked for both Prior Year and Current Year
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2.7 Data Dictionary Notes

- **Field names in bold** represent primary key(s) in their respective tables
 - **Size column** represents maximum field length for Text and Decimal fields
 - Fields without size notation are fixed or variable based on actual values
 - Most text fields are case-sensitive as recorded in the assessor system
-

2.8 Access & Contact

For additional information, data downloads, or to request specific datasets:

- **Source:** Pierce County Assessor-Treasurer Office
- **Website:** <https://www.piercecountywa.gov/>
- **Records Request:** Pierce County Public Records Request Center

3 Summary

In summary, this book has no content whatsoever.

References

Knuth, Donald E. 1984. “Literate Programming.” *Comput. J.* (USA) 27 (2): 97–111.
<https://doi.org/10.1093/comjnl/27.2.97>.